

Autonomous Driving Agent Using Deep Reinforcement Learning: Navigation in Dynamic Urban Environments

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GitHub: https://github.com/22adi66/MetaDrive_RL/tree/main
Drive: <https://drive.google.com/file/d/1Y-nFP4IskUFpznQWSbKTYtD18z0AMxWr/view?usp=sharing>

Abstract—This paper presents the design and implementation of a traffic-aware autonomous driving agent trained using the Soft Actor-Critic (SAC) reinforcement learning algorithm on the MetaDrive simulator. The agent employs a novel 11-component reward function optimizing for speed control, lane keeping, turn quality, traffic distance maintenance, and overtaking detection. A 5-phase curriculum learning strategy progressively increases traffic density from 0% to 45% over 4 million training steps. The system is validated across multiple procedurally generated road types including roundabouts, intersections, curves, and straight roads. Results demonstrate that the agent achieves optimal performance (reward 499) in low-traffic conditions (0–15%) and adapts to dense 45% traffic achieving 33% of baseline performance (reward 165), revealing critical insights about the non-linear difficulty scaling in autonomous driving scenarios. Training was completed in 9.78 hours on CPU infrastructure, processing 20,576 episodes with 518,664 model parameters.

Index Terms—Reinforcement Learning, Autonomous Driving, Soft Actor-Critic, Curriculum Learning, MetaDrive, Traffic Navigation, Reward Shaping

I. INTRODUCTION

Autonomous driving remains one of the most challenging applications of artificial intelligence, requiring real-time decision-making in dynamic, safety-critical environments. Traditional rule-based approaches struggle with the combinatorial complexity of traffic interactions, motivating the adoption of deep reinforcement learning (DRL) methods that can learn adaptive policies from experience.

This work addresses the problem of training an autonomous driving agent capable of navigating complex road scenarios with varying traffic densities. The key challenges include: (1) designing a reward function that balances multiple competing objectives such as speed, safety, and comfort; (2) enabling the agent to generalise across diverse road topologies; and (3) progressively introducing traffic complexity without catastrophic forgetting.

We propose a system built on the MetaDrive simulator (v0.4.3) using the Soft Actor-Critic (SAC) algorithm with an 11-component composite reward function and a 5-phase curriculum learning schedule. The agent is trained on procedurally generated maps including roundabouts, intersections, curves, and straight roads, with traffic density gradually increasing from 0% to 45%.

The primary contributions of this work are:

- A comprehensive 11-component reward function that jointly optimises speed control, lane keeping, turning quality, traffic awareness, and overtaking behaviour.
- A curriculum-based training strategy enabling stable learning progression from traffic-free environments to dense urban conditions.
- Detailed empirical analysis of performance degradation patterns at high traffic densities, revealing a critical 30% traffic “wall.”
- A modular, extensible codebase supporting parallel training, checkpoint management, and video recording.

II. RELATED WORK

Reinforcement Learning for Autonomous Driving. DRL has been extensively applied to autonomous driving, with approaches ranging from Deep Q-Networks (DQN) for discrete action spaces to continuous-control methods like DDPG, TD3, and SAC. SAC, proposed by Haarnoja et al. [1], is particularly suited for autonomous driving due to its maximum entropy framework, which encourages exploration while maintaining stable convergence.

Curriculum Learning. Bengio et al. [2] introduced curriculum learning as a training strategy where tasks are presented in order of increasing difficulty. In autonomous driving, this has been applied to progressively introduce traffic complexity, adverse weather, and challenging road geometries.

Simulation Environments. MetaDrive [3] provides a lightweight, procedurally generated driving simulator with realistic physics, making it suitable for large-scale RL training. Compared to CARLA and SUMO, MetaDrive offers faster simulation speeds and easier programmatic control.

III. METHODOLOGY

A. Environment Design

The agent operates in MetaDrive v0.4.3, a procedurally generated autonomous driving simulator. The environment uses an OXXO map (mixed roundabouts and intersections) with 1 000 procedurally generated scenarios. The physics engine runs at 5 substeps per agent action with a step size of 0.01 seconds and a horizon of 1 000 steps per episode. Crashes are non-terminal to allow learning in traffic.

Observation Space: 144-dimensional sensor fusion vector consisting of: Main LiDAR (72 lasers at 50 m range), Side

detector (20 lasers), Lane detector (20 lasers), and state variables (32 dimensions including velocity and heading).

Action Space: 2-dimensional continuous control — Steering $\in [-1.0, 1.0]$ and Throttle $\in [-1.0, 1.0]$.

B. Reward Function Design

The composite reward function consists of 11 components designed to balance multiple driving objectives:

$$R_{\text{total}} = R_{\text{speed}} + R_{\text{route}} + R_{\text{lat}} + R_{\text{dir}} + R_{\text{smooth}} + R_{\text{yaw}} + R_{\text{border}} + R_{\text{turn}} + R_{\text{traffic}} + R_{\text{overtake}} + R_{\text{stab}} - C_{\text{penalty}} \quad (1)$$

TABLE I
REWARD FUNCTION COMPONENTS AND COEFFICIENTS

Component	Coeff.	Purpose
c_{speed}	1.0	Speed targeting (60 km/h)
c_{route}	1.0	Route progress
c_{lat}	0.5	Lane centre deviation
c_{dir}	0.5	Direction alignment
c_{smooth}	0.2	Action smoothness
c_{yaw}	0.5	Yaw rate control
c_{border}	1.0	Border proximity
c_{turn}	0.8	Turn quality
c_{traffic}	0.5	Traffic spacing
c_{overtake}	1.5	Overtaking bonus
c_{stab}	0.5	Vehicle stability

Speed Control (R_{speed}): Targets 60 km/h with asymmetric overspeeding penalty:

$$R_{\text{speed}} = c_{\text{speed}} \times \left(1.0 - \frac{|v - v_{\text{target}}|}{v_{\text{target}}} \right) \quad (2)$$

Speeds exceeding the target incur an additional $0.3 \times (v - v_{\text{target}})$ penalty.

Penalties: Collision: 10.0 (non-terminal); Off-road: 5.0 (terminal); Time: 0.05/step; Immobility: 1.0 if speed < 0.5 km/h.

C. SAC Algorithm Configuration

TABLE II
SAC HYPERPARAMETERS

Parameter	Value
Policy	MlpPolicy [256, 256]
Learning Rate	3×10^{-4} (Adam)
Replay Buffer	1,000,000
Batch Size	256
Tau (soft update)	0.005
Gamma (discount)	0.99
Entropy Coeff.	Auto (starts 0.861)
Parameters	518,664

D. Curriculum Learning Strategy

Training follows a 5-phase curriculum that progressively increases traffic density to enable stable learning:

TABLE III
CURRICULUM LEARNING SCHEDULE

Phase	Steps	Traffic	Objective
0	0–500K	0%	Baseline driving
1	500K–1M	5%	Adapt to obstacles
2	1M–1.5M	15%	Complex interactions
3	1.5M–2.5M	30%	Dense urban traffic
4	2.5M–4M	45%	Heavy traffic + overtaking

IV. SYSTEM ARCHITECTURE

The codebase follows a modular architecture: `agent/sac_agent.py` handles checkpoint management and model creation; `configs/config.py` centralises all hyperparameters; `env/metadrive_env.py` implements the custom 11-component reward; `training/train.py` manages `SubprocVecEnv` and curriculum integration; `training/curriculum.py` controls traffic density ramping.

Training uses 4 parallel environments via `SubprocVecEnv` for improved sample efficiency. The `CurriculumScheduler` monitors global timesteps and dynamically updates traffic density across all environments. Checkpoints are saved every 100,000 steps with TensorBoard logging.

V. EXPERIMENTAL RESULTS

A. Training Overview

Total timesteps: 4,000,000 | Episodes: 20,576 | Duration: 9h 47m
Updates: 997,338 | Hardware: CPU-only | FPS: 245 (initial) $\rightarrow$ 114 (final)

B. Phase-by-Phase Performance

TABLE IV
PERFORMANCE METRICS ACROSS TRAINING PHASES

Phase	Traffic	Reward	Ep. Len.	Critic Loss
0 (500K)	0%	499	1000	0.010
1 (1M)	5%	499	1000	0.007
2 (1.5M)	15–30%	499 $\rightarrow$ 51	1000 $\rightarrow$ 140	0.007 $\rightarrow$ 47.5
3 (2.5M)	30%	51 $\rightarrow$ 96	140 $\rightarrow$ 172	47.5 $\rightarrow$ 1.43
4 (4M)	45%	96 $\rightarrow$ 165	172 $\rightarrow$ 242	1.43 $\rightarrow$ 2.26

C. Episode Reward vs. Training Steps

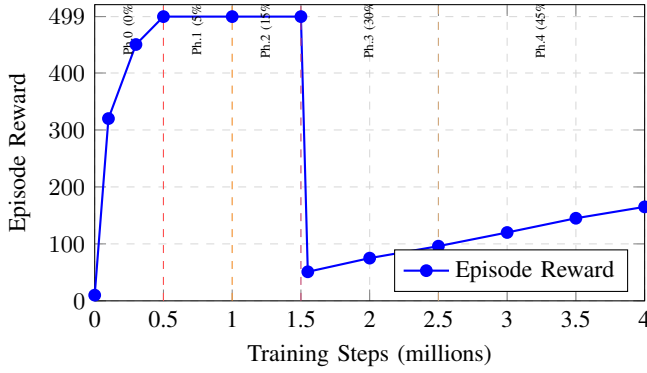


Fig. 1. Episode reward across 4M training steps with curriculum phase transitions marked. The sharp drop at 1.5M steps corresponds to the Phase 3 traffic wall at 30% density.

D. Performance Across Traffic Densities

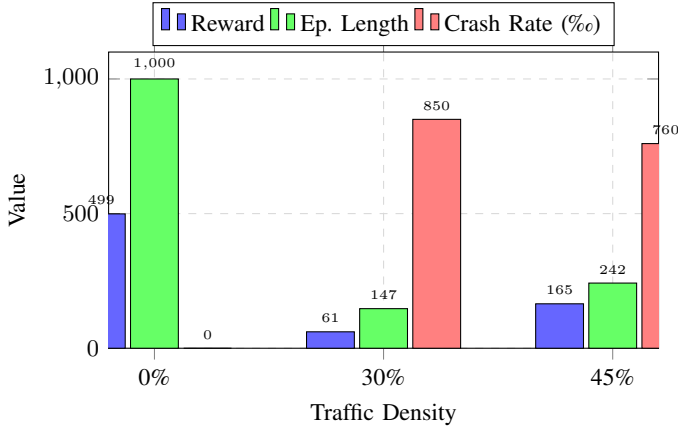


Fig. 2. Performance comparison across traffic densities. Crash rate is shown scaled ($\times 10$) for visibility. At 0% traffic the crash rate is 0.

E. FPS Degradation Across Phases

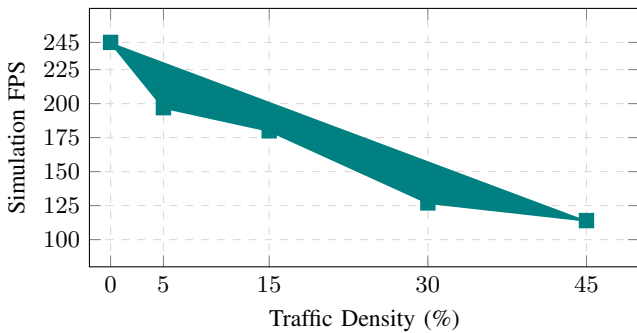


Fig. 3. Simulation FPS degradation as traffic density increases. A 53.5% drop is observed from 0% to 45% traffic density.

F. FPS Degradation Table

TABLE V
TRAINING SPEED ACROSS PHASES

Phase (Traffic)	FPS	Degradation
0 (0%)	245	Baseline
1 (5%)	197	−19.6%
2 (15%)	180	−26.5%
3 (30%)	127	−48.2%
4 (45%)	114	−53.5%

G. Critical Phase 3 Transition

The transition to 30% traffic density at step 1.5M revealed a catastrophic performance collapse:

- Episode reward dropped 87.6% ($499 \rightarrow 51$)
- Episode length dropped 85.3% ($1000 \rightarrow 140$ steps)
- Actor loss plummeted from -49.4 to -4.96 ($10\times$ worse)
- Critic loss exploded from 0.0071 to 47.5 ($6,650\times$ worse)

This “traffic wall” represents a non-linear difficulty spike not captured by linear interpolation of difficulty.

H. Loss Dynamics

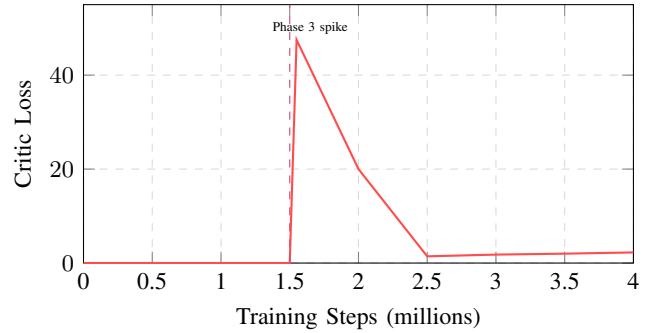


Fig. 4. Critic loss across training. The massive spike at 1.5M steps (Phase 3 transition to 30% traffic) indicates a sharp increase in value estimation difficulty.

I. Performance Ceiling Analysis

TABLE VI
PERFORMANCE COMPARISON ACROSS TRAFFIC DENSITIES

Metric	0% Traffic	30% Traffic	45% Traffic
Reward	499	61 (12%)	165 (33%)
Episode Length	1000	147 (15%)	242 (24%)
Crash Rate	0%	~85%	~76%
Survival Rate	100%	15%	24%

VI. DISCUSSION

A. Strengths

The system demonstrates: (1) rapid baseline convergence with actor loss reaching -41.5 by 100K steps; (2) stable low-traffic performance maintaining optimal reward (499) across 0–15% densities; (3) recovery capability from Phase 3 collapse, improving from reward 96 to 165 over 1.5M steps in

45% traffic; and (4) robust training completion of 4M steps without crashes or numerical instability.

B. Limitations

Key limitations include:

- 1) The 30% traffic wall causing catastrophic collapse, suggesting the 15% $\rightarrow$ 30% curriculum step is too aggressive.
- 2) Early entropy collapse to 0.001 within 200K steps, limiting exploration.
- 3) Insufficient training duration — at 4M steps the agent achieves only 33% of baseline in 45% traffic.
- 4) A 53.5% FPS degradation from replay buffer capacity and traffic simulation overhead.

C. Recommended Improvements

Future work should address:

- 1) **Entropy scheduling:** start at 0.2 and decay to 0.05.
- 2) **Smoother curriculum:** add intermediate 20% and 25% phases.
- 3) **Network scaling:** expand from [256, 256] to [512, 512].
- 4) **Extended training:** 6M–8M steps.
- 5) **Reward rebalancing:** prioritise safety over speed in dense traffic.

VII. CONCLUSION

This work presents a comprehensive deep reinforcement learning system for traffic-aware autonomous driving using SAC with curriculum learning on MetaDrive. The 11-component reward function successfully balances competing driving objectives, and the 5-phase curriculum enables progressive adaptation to increasing traffic complexity.

Key findings include: (1) SAC achieves optimal performance in low-traffic environments within 500K steps; (2) a critical performance cliff exists at 30% traffic density where reward drops 87.6%; (3) the agent demonstrates recovery capability in extreme 45% traffic, improving from reward 96 to 165 over 1.5M steps; and (4) early entropy collapse limits exploration diversity in challenging phases.

Future work should focus on entropy scheduling, smoother curriculum transitions, larger network architectures, and extended training durations to achieve production-ready traffic handling capability.

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Example Driving Scenarios

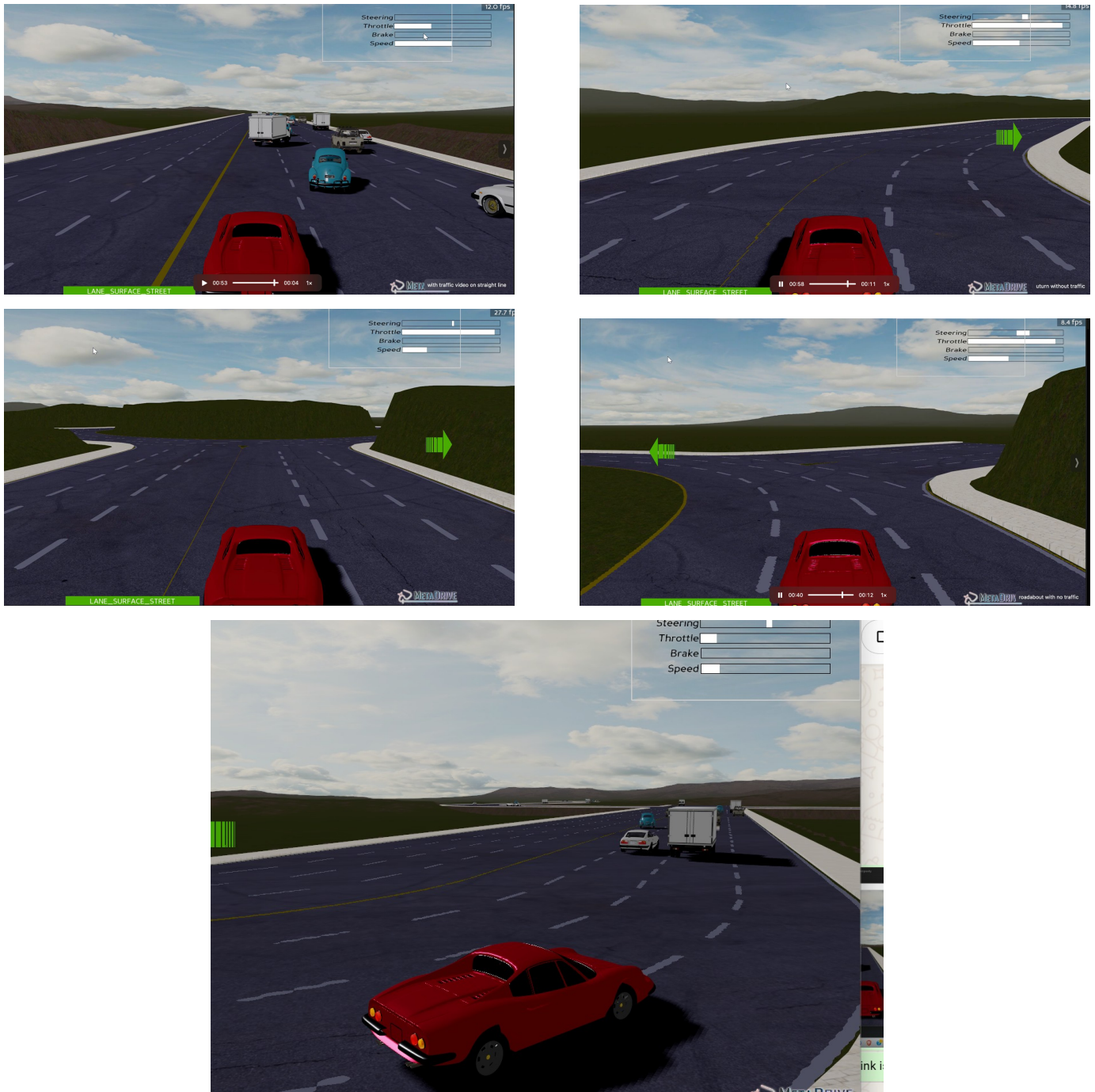


Fig. 5. Driving scenarios in MetaDrive including straight roads, traffic navigation, U-turns, and roundabouts.